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Top Skills

Project Management

Project Planning

Google Apps Script

Certifications

AI for Organizational Leaders by
Microsoft and LinkedIn

Career Essentials in Generative AI
by Microsoft and LinkedIn

Leading Responsible AI in
Organizations

What Is Generative AI?

Generative AI: The Evolution of
Thoughtful Online Search

Vigneshwaran Pugazhenth

Principal R&D Technologist | Co-founder & Managing Trustee |
Entrepreneur | Robotic Process Automation (RPA) & AI Automation |
Virtual Try-On | 3D Virtual Fashion Specialist | Digital Transformation
| R&D Tech Specialist

Bengaluru, Karnataka, India

Summary

As a Principal R&D Technologist, I bring over 11 years of expertise in AI-driven automation, 3D Virtual Fashion, and Augmented Reality (AR/VR) solutions. My focus lies in transforming traditional fashion workflows into cutting-edge digital experiences, streamlining production, and enhancing user engagement through immersive technologies. I specialize in leading complex projects, optimizing digital asset pipelines, and driving industry-relevant digital transformations in fashion.

Key Accomplishments:

3D Virtual Garment Transformation: Spearheaded the transition from traditional photoshoots to 3D Virtual Renders, delivering a 99.7% approval rating while reducing production costs and turnaround time.

AI & Automation for Fashion: Designed and implemented AI-powered automation scripts to enhance garment rendering, texture enhancements, and digital staging, improving overall workflow efficiency.

Cross-Platform 3D Apparel Integration: Led successful collaboration efforts to ensure 3D garment assets are compatible across AR/VR platforms (Snapchat, Meta, Google AR) and e-commerce applications.

Leadership in Innovation: Recognized for my cross-functional leadership and ability to drive AI, RPA, and machine learning solutions in digital fashion, optimizing entire garment production pipelines.

Pioneer in Digital Fashion & Workflow Automation: Innovated AI-driven solutions that minimize manual tasks, reducing labor-intensive work by 70% and enabling real-time garment simulations and transformations.

Additionally, as the Co-Founder & Trustee of Integrity People Trust, I am passionate about creating opportunities for underprivileged communities. We bridge the education and skill-building gap by offering mentorship, financial aid, and workshops on AI and digital transformation. This work is deeply meaningful to me, as I believe in empowering youth with the tools and knowledge to thrive in the digital age.

Passionate about leveraging emerging technologies to redefine the future of fashion and e-commerce, while giving back to the community through digital education and career development initiatives.

Experience

Avataar

10 years 1 month

Principal R&D Technologist

October 2024 - Present (1 year 7 months)

Bengaluru, Karnataka, India

- Led transition from physical garment photoshoots to a 3D rendering pipeline, achieving 99.7% approval rate and reducing production time and costs.
- Designed a scalable 3D asset pipeline for e-commerce, marketing, and AR/VR use cases.
- Developed AI-driven automation scripts for rendering, texturing, and staging, improving efficiency and reducing manual effort.
- Established and automated Meta QC processes, ensuring consistent, high-quality outputs at scale.
- Standardized 3D workflows, pipelines, lighting, and ADCP standards for visual consistency.
- Built end-to-end data management systems, covering asset lifecycle from input validation to delivery.
- Created validation scripts to ensure data integrity and reduce production errors.
- Managed file systems, folder structures, and access control for internal and external teams.
- Authored SOPs, documentation, and training programs to enable onboarding and scalability.

- Provided technical troubleshooting, resolving bottlenecks and ensuring smooth production flow.
- Improved client communication workflows, reducing feedback cycles and accelerating delivery.
- Developed internal tools/plugins to enhance artist and production team productivity.
- Ensured compliance with brand guidelines and global e-commerce standards.
- Scaled pipeline for high-volume production while maintaining quality benchmarks.
- Introduced OneTracker dashboards to monitor pipeline stages, QC pass rates, and turnaround time.
- Collaborated with design, photography, and e-commerce teams to align outputs with business goals.
- Defined R&D roadmap and OKRs, driving innovation and process improvement.
- Delivered for global brands including Victoria's Secret and Patagonia, ensuring premium quality.

Senior Technical Executive

December 2020 - October 2024 (3 years 11 months)

Bengaluru, Karnataka, India

R&D Project 7: Tech with Fashion.

Goal - Virtual Try-On Technology using algorithms and real-time body tracking.

- Redefined Virtual Try-On Technology with advanced garment draping algorithms & real-time body tracking.
- Optimized low-poly 3D apparel models for AR/VR environments, enhancing rendering speeds by 40%.
- Built AI-driven RPA solutions for automated garment staging & QC compliance, ensuring industry-standard outputs.
- Led machine learning research on digital fabric behavior, improving garment movement accuracy in simulations.
- Developed Python-based AI tools for 3D garment visualization & simulation, reducing processing times by 60%.
- Led AI-powered Photoshop automation, improving garment color modifications & design variations for e-commerce.
- Expanded cross-platform feasibility for 3D apparel assets, ensuring compatibility with AR/VR platforms like Snapchat, Meta, and Google AR.

- Spearheaded workflow automation, optimizing garment rendering pipelines & reducing manual intervention by 70%.

Other Key Contributions:

- End-to-End RPA Integration: Built automation frameworks for apparel creation, staging, and delivery, integrating Power Automate & Python with Clo3D.
- Virtual Try-On Innovations: Developed Snapchat AR & full-body tracking solutions, optimizing interactive digital fashion experiences.
- Quality & Compliance Leadership: Implemented AI-driven Quality Control (QC) processes, reducing manual error rates by 80%.
- RPA Expansion: Scaled automation into material creation & 2D CAD workflows, improving digital fashion production cycles.
- Enterprise-Level AI Workflows: Designed automated dashboards & real-time tracking tools for data-driven decision-making.

Technical Executive

April 2016 - December 2020 (4 years 9 months)

Bengaluru, Karnataka, India

3D Fashion & AR/VR Implementation

Project 2: Myntra Phase 1 (Dec 2017 – Feb 2018)

Project 3: Samsung Phase 1 (Mar 2018 – Apr 2018)

Project 4: Myntra Phase 2 (May 2018 – Jun 2018)

Project 5: Google (Mar 2018 – Jan 2019)

Goal: Implementing AR/VR technology in Google-Fashion.

- Created 2D patterns & grading in CAD for fashion applications.
- Transformed 2D patterns into high-quality 3D objects for virtual experiences.
- Applied realistic materials and textures to 3D meshes based on client requirements.
- Optimized low-poly 3D meshes for better performance and reduced file size.
- Recorded animations and rendered high-quality visuals for presentations.
- Simulated apparel on avatars of various sizes for virtual try-ons.

Project 6: Facebook | Instagram | Messenger | Snapchat | Filament | ARCore | ARKit | SceneKit (Jan 2019 – May 2019)

Goal: Enhancing 3D discovery & shopping experiences by integrating AR/VR & AI.

- Led the design and development of interactive 3D experiences tailored to AR/VR platforms.

- Converted 2D fashion designs into hyper-realistic 3D models for e-commerce and social media applications.
- Developed animated 3D assets for AR/VR interactions.
- Rendered high-quality 3D content for integration with Facebook, Instagram, Snapchat, and more.
- Innovated AR/VR-driven fashion experiences for online shopping customers.
- Created avatars, fashion designs, and product visuals to enhance digital shopping engagement.
- Pioneered B2B & B2C research and development, exploring new trends in virtual fashion.
- Contributed to AI-powered shopping solutions, improving personalization and user engagement.

Intern

April 2016 - June 2016 (3 months)

Bengaluru, Karnataka, India

Transformation of 2D Patterns into Realistic 3D Objects

Project 1: Research & Development (Apr 2016 – Nov 2017)

Goal: Finalizing the backend process for a 3D visualization application.

- Developed custom avatars based on client specifications.
- Designed an efficient workflow and pipeline for seamless 3D object creation.
- Conducted in-depth research on realistic 3D object creation, mix & match techniques, and material optimization to enhance output quality while minimizing file size.
- Explored low-poly 3D formats to achieve high-quality visuals with optimized performance.
- Implemented deep learning strategies to reduce manual efforts and enhance automation.
- Streamlined quality control (QC) processes to ensure 100% issue-free deliveries.
- Analyzed stage-by-stage optimization techniques to elevate customer satisfaction.
- Studied market trends and continuously upgraded technology to stay ahead of industry standards.
- Developed automation processes to reduce repetitive manual tasks.
- Applied advanced texturing techniques to achieve a hyper-realistic look for 3D models.

Integrity People Trust

Co-founder & Managing Trustee

August 2019 - Present (6 years 9 months)

Chennai, Tamil Nadu, India

Mission: Bridging the gap in education & skill-building for underprivileged communities.

Impact Highlights:

- Established a mentorship program, connecting professionals with students for career guidance & digital literacy.
 - Sponsored education & provided financial aid to students in need.
 - Conducted workshops on AI & digital transformation, empowering youth with industry-relevant skills.
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